
Supplementary information

An operational perturbation proteomics-based virtual cell model

In the format provided by the
authors and unedited

Supplementary information

Supplementary table legends

Table S1 Cell line and perturbation compound information

A. Cell line annotations used in this study, including source, catalog number, cancer subtype, gene mutations and proliferation rates measured over 48 h log-phase culture.

B. Perturbation annotations used in this study, including the perturbation ID, drug name, catalog number, CAS number, SMILES structure, known molecular targets, associated pathways, and drug subtypes. The final column provides target protein names and their corresponding UniProt IDs. The “Target” column reports target annotations from the drug datasheets, which in some cases refer to broad target families rather than individual proteins. To provide protein-level annotations, we manually curated the corresponding drug target proteins and UniProt IDs from DrugBank (<https://go.drugbank.com>); these are listed in the final target protein/UniProt ID columns. Drug_type1 and drug_type2 are DrugBank-based classifications, with Drug_type1 indicating the broad therapeutic or mechanistic class and Drug_type2 indicating the more specific target or mechanism class.

C. This table summarizes the cell viability assays used to determine drug efficacy based on the 24 h response. Drug responses were labelled as resistant (N) or sensitive (Y) according to the efficacy criteria described in the Methods. Each cell viability assay was performed with three biological replicates. Detailed criteria for defining drug efficacy are provided in the Methods. Compound #3 was available only in limited quantity; therefore, its drug efficacy was not measured, and it was excluded from subsequent model training and evaluation.

Table S2 Sample metadata and proteomic profiles for mtPTDS

A. Sample-level metadata for 2487 non-pooled mtPTDS proteomic samples from the mtPTDS dataset, collected across 11 perturbation times (0, 2, 4, 6, 8, 10, 12, 24, 36, 48, and 60 h) in three TNBC cell lines (HCC1806, HCC1143 and MDA-MB-453-1). Metadata include sample ID, cell line, cell plate ID, perturbation time, treatment category, drug or combination ID, anchor and library drug ID and doses, MS instrument and replicate type. Drug identifiers are indexed in Table S1B. The table includes 2354 independent samples, 84 technical replicates and 49 biological replicates; pooled QC samples are not included in this table. Technical replicates, biological replicates and pooled samples were used for quality control and reproducibility assessment.

B. Matched protein abundance matrix for these 2487 samples. Rows correspond to samples

and columns correspond to 3631 quantified proteins annotated by UniProt accession and gene name concatenated with an underscore (e.g., A0A024RBG1_NUDT4B).

Table S3 PertScore analysis PertScore analysis, pathway enrichment and CDK inhibitor-associated temporal protein dynamics

A. This table presents the PertScore values calculated for each protein within different drug types. The PertScore quantifies how consistently a protein responds to perturbations across multiple conditions, with +1 assigned for upregulation, -1 for downregulation, and 0 for no change. An empirical FDR for PertScore was estimated using a permutation-based approach, as described in the Methods.

B. Differentially regulated proteins at 6, 24 and 48 h after treatment were selected based on PertScore with an empirical FDR < 0.05 and subjected to pathway enrichment analysis using Metascape. The table reports enriched pathways with $P < 0.05$, including drug class, perturbation time, pathway classification, enrichment significance, pathway description and contributing protein hits. Pathways associated with drug mechanism of action (MOA) are shown in Figure 2A.

C. Proteins with contrasting temporal responses between CDK inhibitor-resistant and -sensitive groups. CDK inhibitor-treated samples were stratified by and analysed across 6, 24 and 48 h time points. Proteins were selected if they met at least one of the following criteria: one-way ANOVA for efficacy, $P < 0.05$; one-way ANOVA for time point, $P < 0.05$; or two-way ANOVA for efficacy and time point, $P < 0.05$. The union of proteins passing any of these three criteria was subjected to mFuzz clustering separately in the resistant and sensitive groups using time as the x axis. The resulting clusters were further manually reviewed to retain proteins with clear opposite temporal trends between the two groups. Proteins with opposite temporal trends between the two groups are shown in Figure 2C. Cluster 1 contains proteins downregulated in the resistant group and upregulated in the sensitive group, whereas cluster 2 contains proteins with the opposite pattern.

D. Enrichment analysis was performed using Metascape, and pathways with $P < 0.05$ were retained. Enriched pathways were further curated to emphasize biological functions relevant to the observed protein dynamics, while other disease-related and immune-related pathways were excluded.

Table S4 Beam search of hyperparameters

A. Beam search of hyperparameters for ProteinTalks.

B. Beam search of hyperparameters for other models

Table S5 Performance evaluation and statistical comparison of ProteinTalks and other

models in leave-one-cell line-out cross validation

A. Performance comparison of ProteinTalks and other seven models in leave-one-cell line-out cross validation. At the default classification threshold of 0.5, the scFM baselines classified nearly all samples as belonging to the negative majority class, resulting in nearly identical accuracy values that corresponding to the prevalence of the negative class. Nevertheless, their continuous prediction scores differed, as reflected by their distinct AUROC and AUPRC values.

B. Statistical comparison of ProteinTalks against other seven models in leave-one-cell line-out cross validation.

C. Model performance on individual cell lines. At the default classification threshold of 0.5, the scFM baselines classified nearly all samples as belonging to the negative majority class, resulting in nearly identical accuracy values corresponding to the prevalence of the negative class. Nevertheless, their continuous prediction scores differed, as reflected by their distinct AUROC and AUPRC values.

Table S6 Performance evaluation and statistical comparison of ProteinTalks and other models for leave-one-drug-out validation

A. AUPRC, AUROC, and accuracy of different models tested models in leave-one-drug-out cross validation. At the default classification threshold of 0.5, all scFM baselines classified the samples as the negative majority class (resistant, N), resulting in identical accuracy values corresponding to the negative-class prevalence. Their different AUROC and AUPRC values indicate that the underlying continuous prediction scores were not identical.

B. Comparison of significant differences between ProteinTalks and other models.

C. Model performance on individual drugs. At the default classification threshold of 0.5, all scFM baselines classified the samples as the negative class, resulting in identical accuracy values corresponding to the negative-class prevalence. Nevertheless, differences in their AUROC and AUPRC values for some drugs indicate that their underlying continuous prediction scores were not identical.

Table S7 Compound information, efficacy labels and model evaluation on the PTNC dataset

A. Information on the 81 compounds included in the PTNC dataset. Compounds were indexed by their corresponding Pert_IDs and were purchased from TargetMol. The table includes compound name, catalog number, CAS number, SMILES, molecular formula, pathway annotations, indication, solvent, target annotations and drug type.

B. This table summarizes the cell viability assays used to determine compound efficacy based on the 24 h response. Each cell viability assay was performed with three biological replicates. Drug responses were labelled as resistant (N; negative class) or sensitive (Y; positive class) according to the efficacy criteria described in the Methods. Cell-line-compound combinations with inconsistent labels across repeated experiments were removed; therefore, the resulting compound-by-cell-line matrix is not fully orthogonal.

C. AUPRC, AUROC, and accuracy of different models tested on the PTNC dataset. For the scFM baselines, predictions from multiple random-seed runs showed near-constant outputs across samples (prediction collapse). Accordingly, their AUPRC values were close to the positive-class prevalence baseline in this dataset.

D. Comparison of significant differences between ProteinTalks and other models.

Table S8 Sample metadata, compound information and proteomic profiles for PTPC.

A. Sample-level metadata for the PTPC dataset, an independent perturbation proteomics dataset generated across five cancer cell lines (A375, HCT116, NCI-H1975, NCI-H2170 and Panc 10.05). Proteomic profiles were collected at 6 h and 24 h after treatment, with control samples assigned a perturbation time of 0 h. This table lists 2217 non-pooled samples, comprising 2021 independent proteomic samples and 196 technical or biological replicates. The 90 pooled samples are included in the total PTPC sample count but are not listed in this table. After accounting for pooled and replicate samples, PTPC contained 2021 independent proteomic samples for model evaluation.

B. Compound information for the 716 perturbation identifiers used in PTPC, corresponding to 708 unique compound names. All compounds were purchased from TargetMol. Information includes perturbation ID, catalog number, compound name, CAS number, SMILES, molecular formula, pathway annotations, indication, solvent, target annotations and compound type.

C. Matched protein abundance matrix for the same 2217 non-pooled sample records. Rows correspond to samples and columns correspond to quantified proteins annotated by UniProt accession and gene name concatenated with an underscore (e.g., A0A1B0GUA7_TEX51).

Table S9 Performance evaluation and statistical comparison of ProteinTalks and other models on the PTPC dataset

A. AUPRC, AUROC, and accuracy of different models tested on the PTPC dataset. At the default classification threshold of 0.5, the scFM baselines classified nearly all samples as the negative majority class, resulting in identical accuracy values that matched the prevalence of

the negative class. However, the differences in AUROC and AUPRC indicate that the underlying continuous prediction scores were not identical.

B. Comparison of significant differences between ProteinTalks and other models.

Table S10 Cytotoxicity assays for predicted drug combinations

Results from cytotoxicity assays for drug combinations predicted by the ProteinTalks model to be synergistic, including combinations of bosutinib with tucatinib in HCC1395 cells, bosutinib with tucatinib in HCC1143 cells, and bosutinib with abemaciclib in HCC70 and HCC1806 cells. The treatments were administered as follows: HCC1143 cells received a combination of bosutinib and tucatinib with a fixed concentration of 1 μ M bosutinib; and HCC70 and HCC1806 cells were exposed to a combination of abemaciclib and bosutinib with a fixed concentration of 1 μ M bosutinib. Each assay plate included triplicate wells. The shifts in efficacy (Δ Emax), representing the reduced cell viability, were determined by calculating the difference in efficacy between the observed combination response and the expected response based on Bliss independence (Bliss, C. I. , 1939, *Annals of applied biology* **26**, 585-615).

Table S11 SHAP value-based protein ranking, cancer association, and functional annotation

A. Protein ranking by SHAP-value in predicting chemotherapy drugs' sensitivity. B.

Literature-curated cancer associations of the top 100 SHAP-ranked proteins. This sheet summarizes literature-supported functional evidence linking the top 100 proteins ranked by SHAP values to breast cancer or other cancer types, with particular emphasis on functional studies in non-breast cancer contexts rather than associations from large-scale datasets alone.

C. Biological process GO annotation of 18 proteins with limited functional characterization in cancer.

Table S12 Evaluation of transfer learning and drug efficacy prediction in PDX models

A. Evaluation of drug efficacy prediction performance after pretraining with different proportions of PTDS-1 perturbation proteomic data followed by refinement on a subset of transcriptomic data from PDX models. The remaining PDX transcriptomic data were used for model evaluation. Model-1-wo was trained from scratch using only the same subset of PDX transcriptomic data, without transfer learning from perturbation proteomic pretraining. Models labelled by percentage indicate the proportion of PTDS-1 perturbation proteomic data used during the first-stage pretraining. Statistical significance was assessed using the Mann-Whitney U test.

B. AUPRC and AUROC of different models tested on the PDX-2015nm dataset. Given the

strong class imbalance in this dataset, model comparison focused on threshold-independent ranking metrics rather than accuracy.

C. Statistical comparison of model performance on the PDX-2015nm dataset using the same evaluation results reported in Table S12B. Pairwise differences between ProteinTalks and each comparator model were assessed for AUPRC and AUROC.

Table S13 Clinical characteristics and proteomic profiles of 501 TNBC patients from the TNBC-HMU

A. Clinical and treatment metadata for 501 patients with triple-negative breast cancer from the TNBC-HMU cohort. Metadata include patient ID, age, TNM stage, recurrence status, recurrence-free survival time, survival status, overall survival time and treatment drug combinations. Drug identifiers are indexed in Table S1B.

B. Matched FFPE tumor proteomic abundance matrix for the same 501 patients. Rows correspond to patients and columns correspond to 3583 quantified proteins annotated by UniProt accession and gene name concatenated with an underscore (e.g. A0A1B0GUA7_TEX51), with patient IDs provided for sample matching.

Table S14 Drug library information and baseline proteomic profiles of PDO patients

A. Information for the 3000-compound repurposing drug library used for PDO response prediction. Each compound is annotated with compound name, catalog number, CAS number, SMILES, molecular formula, clinical or disease-related condition, indication, pathway annotation, target information, bioactivity, reference, and solvent. All compounds were purchased from TargetMol.

B. Baseline proteomic profiles of three PDO patient samples. Rows correspond to quantified proteins annotated by Uniprot accession, gene names and columns correspond to baseline protein abundance measurements for each PDO patient sample.

Table S15 Prognostic relevance of PALLD, RCOR3, and ZFYVE19 in breast cancer patients

BCa, breast cancer; TNBC, triple-negative breast cancer; IDC, invasive ductal carcinoma; BCa+ chemo, breast cancer with receipt of chemotherapy; BCa+chemo+radio, breast cancer with receipt of chemotherapy and radiotherapy; DFS, disease-free survival; OS, overall survival; RFS, recurrence-free survival, DMFS, distant metastasis-free survival; HR, hazard ratio; CI, confidence interval.

※ P-value was derived from Kaplan–Meier survival analysis (log-rank test). All other p-values were calculated by univariate Cox proportional hazards regression based on individual

protein expression levels

Table S16 Concentration design of drug combinations used in the synergy assay

Table S17 Synergy of drug combinations in knockdown and wildtype cell lines

Dose-response curves for docetaxel combined with a fixed concentration of carboplatin (10 μ M or 40 μ M) in the PALLD-knockdown (PALLD group) and the control (NC group) HCC38 cells. The table summarizes the Emax values derived from the fitted dose-response curves for each treatment condition. Each experimental condition was performed in three independent biological replicates.

Table S18 Mapping of results sections to methods sections

Supplementary results

Rationale for informativeness of four time points

We provide a mathematical interpretation of how observations at baseline and three post-treatment time points constrain the fitted trajectories. Let the proteomic state evolve as $\dot{x}(t) = f_\theta(x(t), u)$, where $x \in R^p$ and u is a time-invariant perturbation representation. For each trajectory we observe $x(0)$ (pre-treatment) and three post-treatment states $x(t_{6h}), x(t_{24h}), x(t_{48h})$. Although sparse in time, this design can provide informative constraints for learning stable and predictive vector field under mild regularity assumptions, for three reasons:

(1) Local flow identification via time-Taylor constraints.

By integrating the ODE, $x(t) = x(0) + \int_0^t f_\theta(x(s), u) ds$. This local expansion motivates the approximation that, around $t = 0$, the flow admits the series:

$$x(t) = x(0) + tv + \frac{t^2}{2}a + O(t^3), \quad v = f_\theta(x(0), u), \quad a = J_x f_\theta(x(0), u)v.$$

Three distinct $t_k > 0$ provide linearly independent constraints that identify v (constrain an effective early direction of change under a local approximation) and a (local curvature driven by the Jacobian); the truncation error is $O(t^3)$ and is bounded when f_θ is locally Lipschitz with bounded Jacobian. Thus, with three post-baseline points, the model are constrained by the short-to-intermediate-horizon behavior of dynamics in a local quadratic approximation.

(2) Global constraints across initial conditions and controls.

To further enhance the generalization capability and identifiability of the model, in ProteinTalks, each trajectory is sparsely sampled, and the dynamical parameters θ are shared across many cell-line baselines and perturbations. For cell line i and drug j , the initial latent state $z_{ij}(0)$ jointly encodes the baseline proteomic profile and perturbation descriptor D_j , while D_j also directly conditions the latent dynamics through $f_\theta(z_{ij}(t), t, D_j)$. Training couples all trajectories through this shared vector field and enforces the exact multi-time integral relation:

$$z_{ij}(t_k) = z_{ij}(0) + \int_0^{t_k} f_\theta(z_{ij}(s), s, D_j) ds, \quad k = 1, 2, 3.$$

Across all cell lines i and drugs j , each trajectory adds several time-point constraints. Joint fitting across multiple initial states, perturbations and sampling times provides multiple constraints on a shared parameterization and may reduce the range of models consistent with the observed data, and may further improve practical identifiability of the learned flow θ .

This empirical identifiability depends on appropriate regularization and diverse coverage of initial-state perturbation pairs $(z(0), D)$. In our setting, the dataset provides broad coverage of these conditions, with many cell lines contributing distinct baseline proteomic states and many drugs contributing distinct perturbation controls.

(3) Stability and additional supervision.

Under Picard-Lindelöf conditions, the flow is unique and perturbation-stable by using Grönwall-type bounds, which describe continuous dependence of trajectories on initial states

or inputs over a finite interval. Training at three times jointly constrains the integrated flow over the interval $[0, t_3]$; Moreover, the model is simultaneously supervised to predict the binary efficacy label, which imposes an additional functional constraint on the trajectory, further reducing the admissible hypothesis space for f_θ .

Beyond these ODE-based properties, the neural-ODE parameterization can further encode a smooth, shared vector field that renders the integral constraints well-conditioned and enables stable and efficient adjoint-based optimization with sparse time sampling.

In summary, while denser sampling may further improve identifiability, four time points (baseline plus three points after treatment) can provide informative constraints that capture instantaneous responses and leading nonlinear dynamics. When aggregated across many initial conditions and drug controls, these constraints indeed provide empirical rationale to train a stable neural ODE that generalizes to unseen combinations of cell lines and drugs.

No significant improvement when incorporating embeddings from multiple transcriptomic foundation models into ProteinTalks

To evaluate whether transcriptomics integration could further improve ProteinTalks, we incorporated pretrained transcriptomic foundation models as encoders within the ProteinTalks framework. Briefly, for each sample, the sample-specific baseline proteomic abundance profile was first passed through each transcriptome-pretrained foundation model to generate sample-specific embeddings, which were then input to ProteinTalks. This design integrates transcriptome-pretrained representations with proteomic inputs for dynamic modeling of proteomic perturbation trajectories. The same data and training strategy as in the original ProteinTalks were used: pretraining on PTDS, followed by few-shot learning on PTPC and testing. Among the three transcriptome-pretrained models evaluated, none demonstrated a statistically significant improvement over the original ProteinTalks across either AUPRC or AUROC metrics (**SI Figure 1**), indicating that transcriptome-pretrained embedding integration does not yield a meaningful performance gain relative to the proteomics-only framework under the current experimental conditions.

Biological insights and validation of top 100 proteins prioritized by ProteinTalks

Manual curation of the top 100 proteins prioritized by ProteinTalks revealed that 63 have established roles in breast cancer diagnosis, treatment, or prognosis, and 20 proteins with functional evidence reported mainly in other cancer types (**Table S11B**). A subset of 18 proteins showed limited functional characterization in cancer. Biological process GO annotation of these proteins indicated diverse functions, including cellular component organization, transcription-related processes and zinc-finger protein-associated functions, with the remaining proteins linked to other biological processes (**Table S11C**). Although functional cancer evidence remains limited for this subset, several annotations suggest potential relevance to chemotherapy response. Chemotherapy primarily disrupts proteins governing cellular component organization through its mechanisms of action. For example, SYNE2, which serves as an actin cytoskeleton linker that maintains subcellular spatial organization ¹, could plausibly be perturbed by antimetabolic agents. Proteins linked to nucleic acid metabolic processes are likely affected by alkylating agents ², given their DNA-targeting

activity. Moreover, zinc-finger proteins, the largest class of DNA-binding proteins³, participate in DNA damage recognition and repair—pathways that are critical for resistance to alkylating agents. Collectively, these 18 proteins constitute mechanistically plausible candidates whose perturbation could modulate chemotherapy efficacy.

The prognostic relevance of ZFYVE19, PALLD and RCOR3 was validated across five independent public datasets and our in-house TNBC-HMU cohort, showing a consistent association between their expression levels and differential prognosis in breast cancer patients (**Tables S13, and S15**). Protein expression was also verified at the tissue level in the TNBC-HMU cohort via immunohistochemical (IHC) analysis (**SI Figure 2**).

Among the three identified proteins, ZFYVE19, a cytokinesis checkpoint regulator that delays abscission to prevent DNA damage accumulation⁴, might exhibit altered expression upon exposure to alkylating agents. Palladin (PALLD) is an actin-associated protein essential for the normal organization of the actin cytoskeleton and focal adhesions^{5,6}. REST corepressor 3 (RCOR3) functions as a competitive inhibitor of LSD1 by antagonizing RCOR1/2-facilitated nucleosomal demethylation, thereby suppressing erythro-megakaryocytic differentiation⁷. Given that these top 100 proteins distinguished sensitive from resistant TNBC patients in response to neoadjuvant chemotherapy with docetaxel and carboplatin, we next investigated whether PALLD contributes to the synergistic effect of this drug combination *in vitro* (**SI Figure 3, Table S16**). Using RNA interference to knock down PALLD in HCC38 cells, we observed no significant difference in cell viability under single-agent treatment conditions compared with control cells. However, PALLD knockdown significantly enhanced the synergistic effect of the docetaxel-carboplatin combination (**SI Figure 3, Table S17**), suggesting that PALLD suppression potentiates the cooperative cytotoxic activity of this regimen.

Supplementary methods

Cell line authentication and mycoplasma testing

All cell lines were authenticated by short tandem repeat (STR) profiling, most recently performed on June 12, 2026. Mycoplasma contamination testing was negative, assessed by PCR-based detection using Genetic Testing Biotechnology (GTB)'s Mycoplasma Detection Kit, which detects over 500 mycoplasma species with a detection limit of 0.1 ng template DNA. The method was performed as previously described ⁸. Both STR profiling and mycoplasma testing were conducted by Genetic Testing Biotechnology Corporation (Suzhou, China).

SMILES generation

Using the SMILES representations of the drug molecules, we employed the R-package ChemmineR ⁹. We set the parameters as follows, functions MW and MF parameter addH=FALSE, function bonds parameter type="charge", function groups parameter type="countMA", function rings parameters upper=6, type="count", arom=TRUE. We obtained the 881-dimensional drug molecular fingerprints (DMF) and the 54-dimensional drug physicochemical properties for each drug.

Integration of transcriptomic foundation models as encoders into ProteinTalks

To integrate transcriptomic information into ProteinTalks while preserving proteome-driven dynamical supervision, we converted the model input from a scalar abundance per protein to a feature vector per protein obtained from transcriptome-pretrained foundation model checkpoints. Because these checkpoints were pretrained on large transcriptomic corpora, their embedding spaces encode gene-level statistical structure learned from transcriptome data. By generating an embedding for each protein from these transcriptome-pretrained models, ProteinTalks receives transcriptomics-informed protein features, while the dynamics model and training losses remain defined by time-resolved perturbational proteomics. For each sample, we started from the pre-perturbation proteomics profile and mapped each measured protein to its corresponding gene symbol (when available). We then queried a transcriptome-pretrained checkpoint to obtain an embedding for each gene, yielding a protein-by-feature matrix:

$$X \in \mathbb{R}^{m \times k}$$

where m is the number of measured proteins and k is the embedding dimension of the transcriptome-pretrained foundation model. Each row corresponds to one protein (via its gene symbol) and serves as the sample-specific input feature vector for that protein. We evaluated three transcriptome-pretrained foundation models for feature construction: GeneCompass, Geneformer, and UCE. In all experiments, these models were used strictly as feature extractors (pretrained weights frozen), so that any performance differences reflect the information content of the transcriptome-pretrained embeddings rather than additional end-to-end retraining of the transcriptome-pretrained model.

The transcriptome-pretrained, sample-specific embedding matrix was used at the ProteinTalks input layer in place of the scalar-abundance representation. The downstream architecture, including the perturbation-conditioned dynamics module and prediction heads, and all

optimization hyperparameters followed the standard ProteinTalks configuration. We used the same training strategy as ProteinTalks: pretraining on PTDS, followed by few-shot adaptation on PTPC, and evaluation on the same held-out test protocol. This design integrates transcriptome knowledge through transcriptome-pretrained embeddings while keeping the learning signal and evaluation grounded in perturbational proteomics.

Multi-task loss optimization

ProteinTalks uses two task losses: Loss1 for time-resolved proteome reconstruction and Loss2 for drug-efficacy or synergy prediction. In the analyses reported in this study, these objectives were optimized with the fixed combined loss:

$$Loss = (1 - \lambda)Loss1 + \lambda Loss2, \lambda = 0.8.$$

where lambda is the hyperparameter balancing the two tasks and is held constant during the reported training runs.

As an optional optimization strategy, the same multi-task framework can also mitigate gradient conflicts between Loss1 and Loss2 by calculating the cosine similarity between their gradients:

$$c = \frac{\langle \nabla Loss1, \nabla Loss2 \rangle}{\|\nabla Loss1\| \|\nabla Loss2\|}.$$

Conflict strength is defined as a non-negative quantity:

$$a = \eta \max(0, -c), \eta = 0.01.$$

Here η is the adjustment rate, corresponding to 0.01. When the gradient directions are non-conflicting ($c \geq 0$), the original task weights are retained:

$$w_{Loss1} \leftarrow w_{Loss1}, \quad w_{Loss2} \leftarrow w_{Loss2}.$$

When the gradient directions conflict ($c < 0$), the optional scheme prioritizes the drug-efficacy or synergy prediction task by increasing the weight of Loss2 while decreasing the weight of Loss1:

$$\begin{aligned} w_{Loss1} &\leftarrow \text{clip}(w_{Loss1} - a), \\ w_{Loss2} &\leftarrow \text{clip}(w_{Loss2} + a). \end{aligned}$$

After this update, the task weights are clipped to the range $[0, 1]$ and renormalized so that their sum remains one:

$$w_{Lossk} \leftarrow w_{Lossk} / (w_{Loss1} + w_{Loss2}), \quad k \text{ in } \{1, 2\}.$$

The adaptive multi-task loss can then be written as:

$$Loss_{adaptive} = w_{Loss1} Loss1 + w_{Loss2} Loss2.$$

The resulting weighted loss is backpropagated through the model to update the parameters. This gradient-conflict option was not used for the ProteinTalks results reported in this study, which used the fixed $\lambda = 0.8$ weighting described above.

IHC staining and scoring

Formalin-fixed, paraffin-embedded tissue sections (4 μm) were dewaxed in xylene and rehydrated through a graded ethanol series. Endogenous peroxidase activity was quenched with 3% hydrogen peroxide for 15 min at room temperature. Antigen retrieval was performed by heating sections in either Ethylenediaminetetraacetic acid (EDTA) buffer (pH 9.0) or citrate buffer (pH 6.0) as appropriate for each antibody. Sections were blocked with goat serum for 40 min and incubated overnight at 4°C with the following primary antibodies: anti-ZFYVE19 (1:100, 23163-1-AP, Proteintech), anti-PALLD (1:200, 10853-1-AP, Proteintech),

and anti-RCOR3 (1:100, AP55371, Abcepta). After washing, sections were incubated with HRP-conjugated goat anti-rabbit IgG polymer (PV-6001, ZSGB-BIO) at 37°C for 30 min, followed by 3,3'-Diaminobenzidine (DAB) chromogen development and hematoxylin counterstaining.

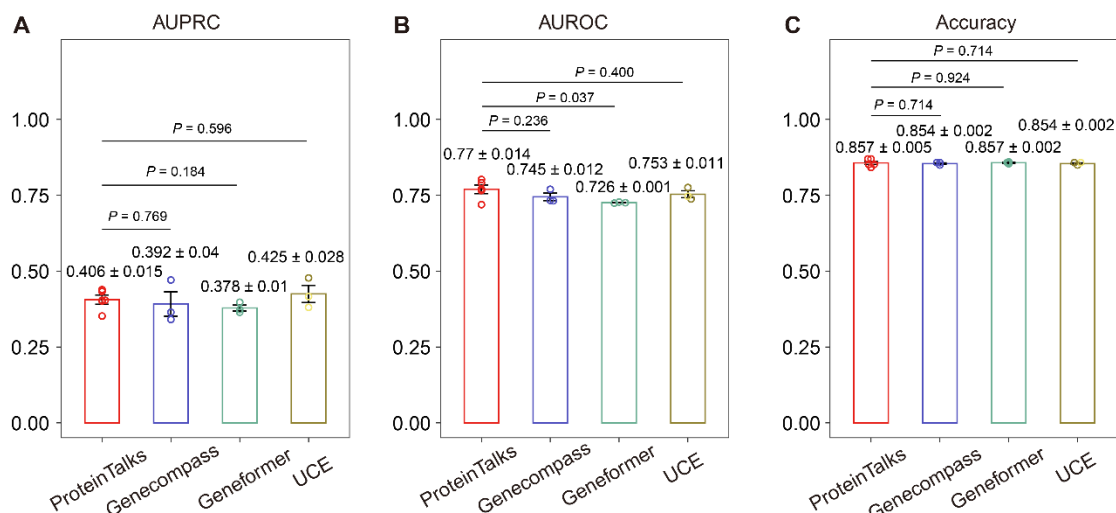
Staining was independently scored by two experienced pathologists blinded to clinical information, using the H-score method¹⁰. Each section was evaluated for staining intensity on a scale of 0–3 (0, negative; 1, weak; 2, moderate; 3, strong) and the percentage of positive tumor cells on a scale of 1–4 (1, ≤25%; 2, 26–50%; 3, 51–75%; 4, >75%). The H-score was calculated as the product of these two parameters, yielding a total score ranging from 0 to 12. Samples were classified as negative (0), low (1–4), moderate (5–8), or high (9–12) expressors.

Dose-response curve fitting and IC₅₀ comparison

Luminescence values were normalized to the untreated control wells and used to calculate relative cell viability. For curve fitting, viability values were first converted to relative inhibition by subtracting them from one, and the transformed values were constrained to the range of 0 to 1 to reduce the impact of out-of-range measurements caused by experimental variability. Dose-response curves were fitted using a sigmoidal model, and IC₅₀ values were estimated from the fitted curves. The fitted values from this model were used for the main dose-response plots and IC₅₀ estimation in **Figures 5G-I**. For each concentration, three independent biological replicates were collected and processed in the same manner; these replicate-level values are shown overlaid on the fitted curves in **SI Figure 4**. Statistical comparisons between cisplatin (M0) and each compound (M1) were performed using likelihood-ratio tests based on the fitted models. The deviance was calculated as $D=2 \times (\log\text{LikM1} - \log\text{LikM0})$. The resulting deviance was compared with a chi-squared distribution, with degrees of freedom equal to the difference in the number of fitted parameters. A *P*-value < 0.05 was considered statistically significant.

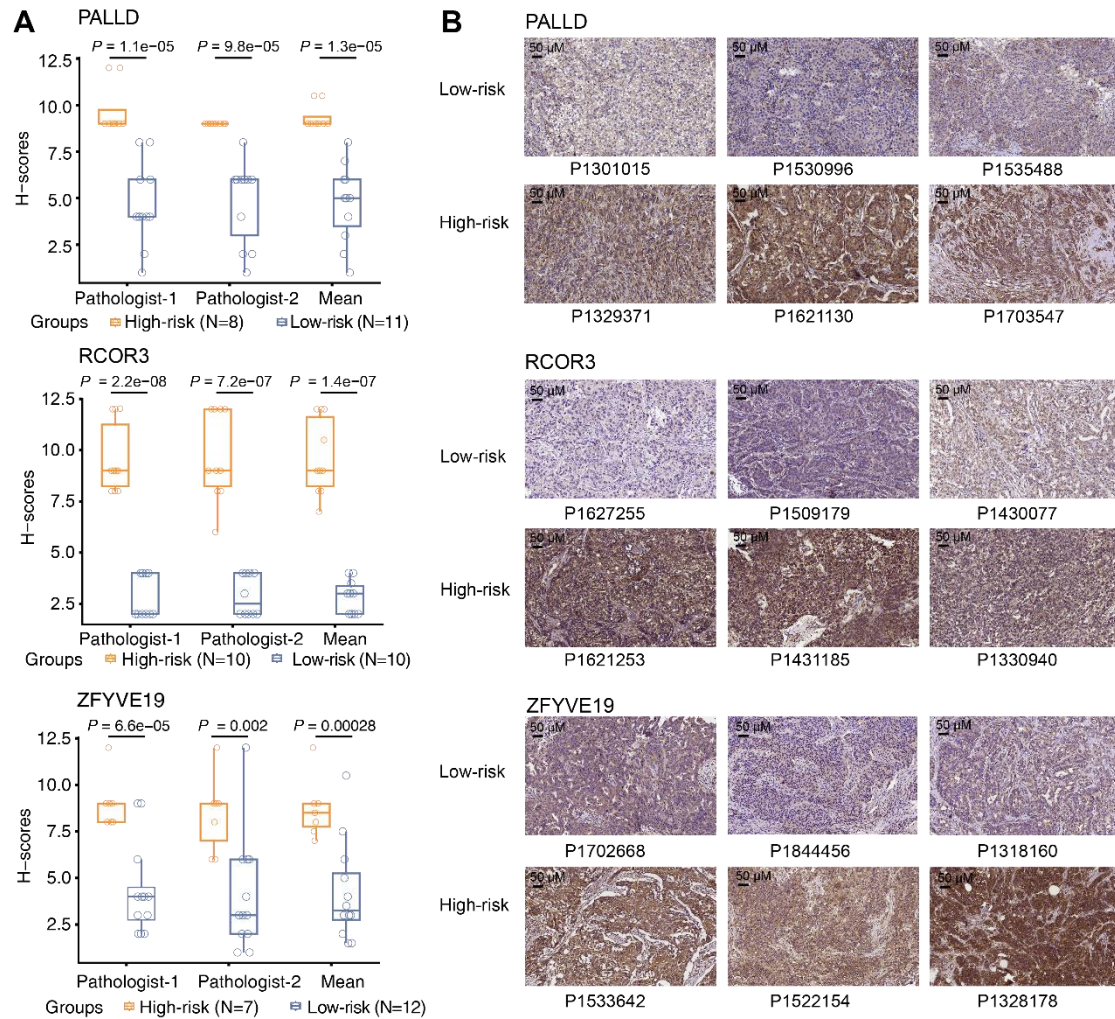
Supplementary information figures

Supplementary information Figure 1



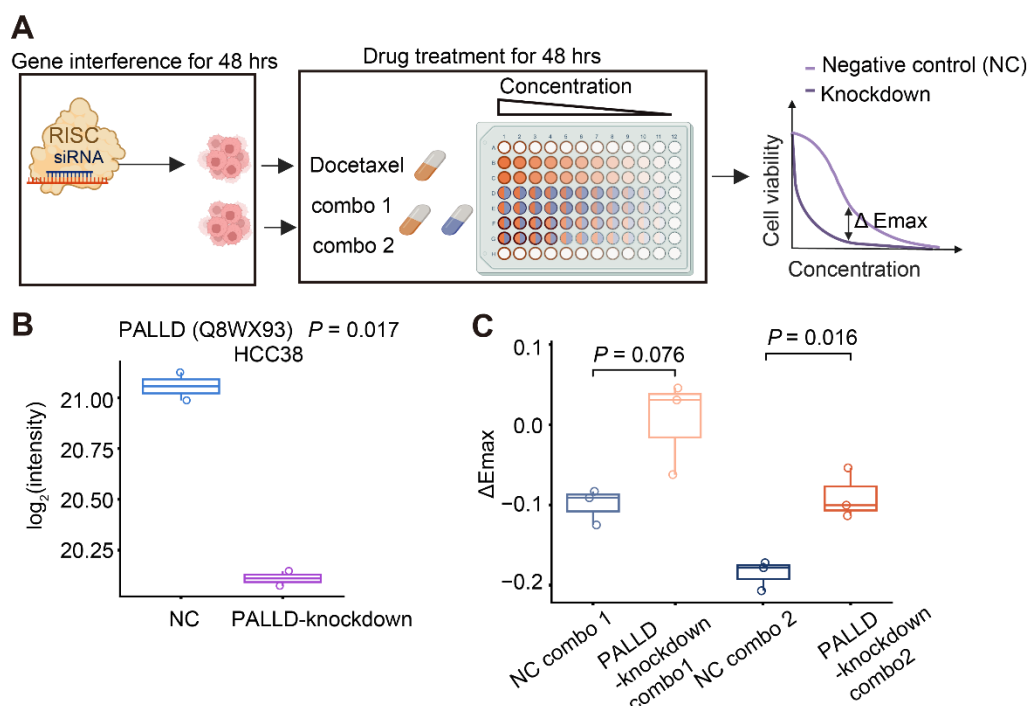
SI Figure 1. Transcriptome-pretrained per-protein features for ProteinTalks and transfer learning across five non-breast cancer cell lines. (A)-(C) Transcriptomics was integrated into ProteinTalks by changing the input representation from one scalar abundance per protein to one transcriptomics-informed feature vector per protein derived from transcriptome-pretrained foundation model checkpoints (GeneCompass (blue), Geneformer (green), or UCE (yellow)). Each measured protein was mapped to its gene symbol and assigned a gene embedding from the transcriptome-pretrained model, yielding a protein-by-feature matrix that served as input to ProteinTalks; the transcriptomic model weights were used as frozen feature extractors. ProteinTalks architecture and training protocol were otherwise unchanged: pretraining on PTDS, followed by few-shot adaptation on PTPC (n=2021 independent proteomic samples) and evaluation using the standard test protocol. Comparative performance of ProteinTalks on five non-breast cancer cell lines in PTPC, including AUPRC (A), AUROC (B), and Accuracy (C). ProteinTalks (n=5 random seeds); GeneCompass, Geneformer, and UCE (n=3 random seeds each). Statistical significance was assessed by two-sided Student's *t*-test and horizontal brackets indicate pairwise comparisons. Error bars represent standard error (s.e.) across random seeds.

Supplementary information Figure 2



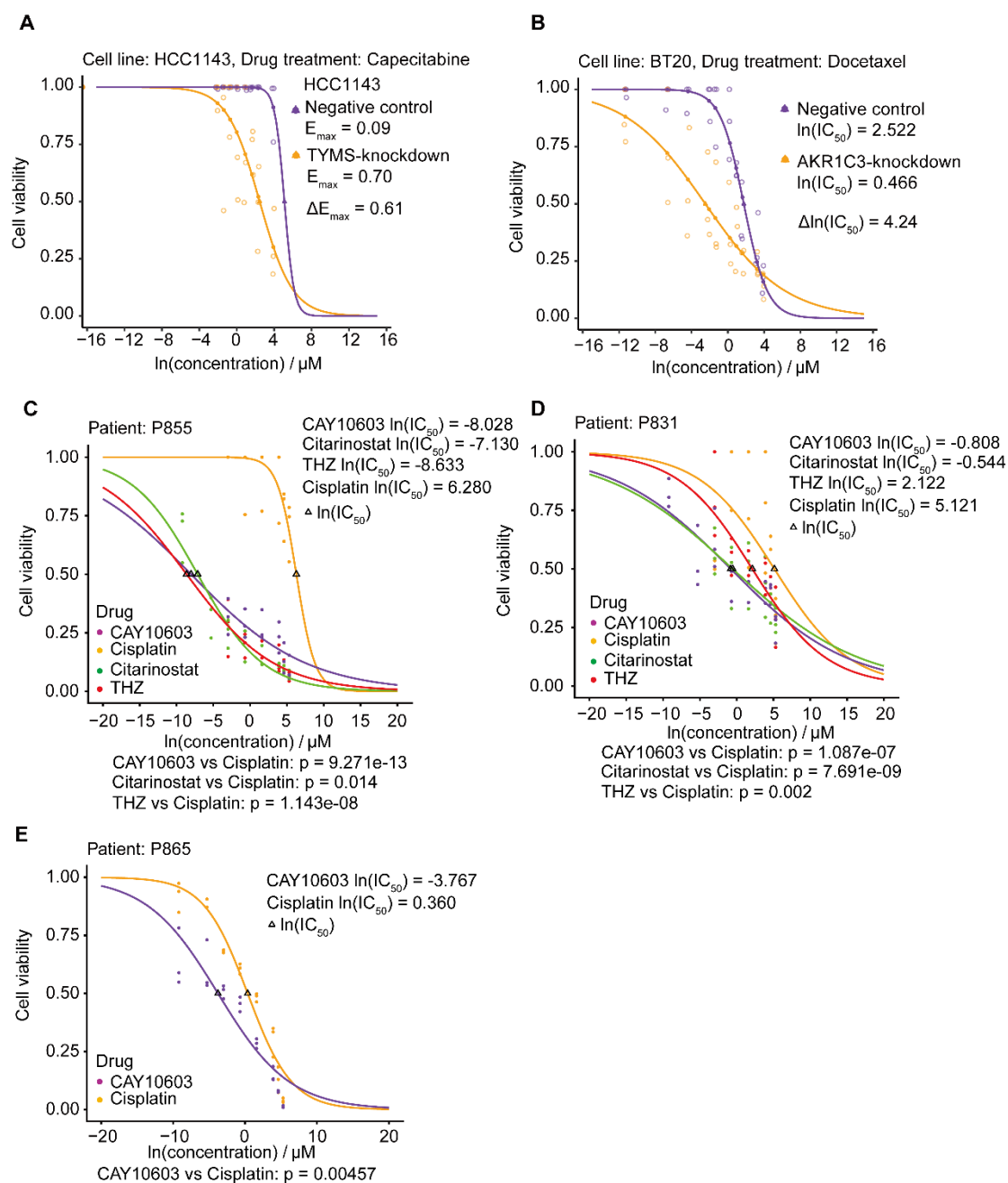
SI Figure 2. IHC validation of PALLD, RCOR3, and ZFYVE19 expression in the TNBC-HMU cohort. (A) Quantification of IHC staining scores for PALLD, RCOR3, and ZFYVE19 in the TNBC-HMU cohort. H-scores assigned by two pathologists (Pathologist 1 and Pathologist 2) and their mean are presented for each case. N indicates the number of patients in each group (high-risk vs. low-risk). Statistical significance between groups was assessed using the two-sided Student's *t*-test. Box plots display median (center line), interquartile range (box), and 1.5× IQR (whiskers). (B) Representative IHC images from three patients per risk group for each of the three proteins. Scale bar, 50 μ m.

Supplementary information Figure 3



SI Figure 3. *In vitro* validation of PALLD biological function in modulating drug synergy. (A) Experimental workflow for drug synergy validation. HCC38 cells were subjected to siRNA-mediated PALLD knockdown, followed by treatment with serial dilutions of docetaxel alone or in combination with carboplatin ($n=3$ biological replicates). Icons were created in BioRender; Guo, T. <https://BioRender.com/rtk4ls8> (2026). (B) Box plot showing $\log_2$ -transformed protein abundance measured by mass spectrometry (MS) to assess RNA interference efficacy. PALLD knockdown efficiency was confirmed in HCC38 cells relative to siRNA control ($n=4$ biological replicates). Statistical significance was assessed by two-sided Student's t -test. Box plots display median (center line), interquartile range (box), and $1.5 \times$ IQR (whiskers). (C) Efficacy shift (ΔE_{max}) quantifying the synergistic effect of the drug combination. ΔE_{max} is defined as the difference in maximum response between the combination treatment and docetaxel monotherapy; a larger ΔE_{max} indicates a stronger synergistic effect. Statistical significance was assessed by two-sided Student's t -test. Box plots display median (center line), interquartile range (box), and $1.5 \times$ IQR (whiskers).

Supplementary information Figure 4



SI Figure 4. Biological replicate measurements overlaid on fitted dose–response curves.

Each point represents one biological replicate after normalization, transformation to relative inhibition, and scaling to the 0–1 range. The fitted curve represents the model-estimated mean response. Thus, whereas Fig. 4D, Extended Data Fig. 3D and Fig. 5G-I show one fitted value per concentration, this supplementary figure displays the replicate-level observations used for model fitting and reproducibility assessment. **(A)** Replicate-level data related to Extended Data Figure 3D. **(B)** Replicate-level data related to Figure 4D. **(C)-(E)** Replicate-level data related to Figures 5G-I, respectively.

References

- 1 Wallis, S. S. *et al.* The ESCRT machinery counteracts Nesprin-2G-mediated mechanical forces during nuclear envelope repair. *Dev Cell* **56**, 3192-3202 e3198 (2021).
- 2 Mullen, N. J. & Singh, P. K. Nucleotide metabolism: a pan-cancer metabolic dependency. *Nat Rev Cancer* **23**, 275-294 (2023).
- 3 Klug, A. The discovery of zinc fingers and their development for practical applications in gene regulation and genome manipulation. *Q Rev Biophys* **43**, 1-21 (2010).
- 4 Thoresen, S. B. *et al.* ANCHR mediates Aurora-B-dependent abscission checkpoint control through retention of VPS4. *Nat Cell Biol* **16**, 550-560 (2014).
- 5 Parast, M. M. & Otey, C. A. Characterization of palladin, a novel protein localized to stress fibers and cell adhesions. *J Cell Biol* **150**, 643-656 (2000).
- 6 Dixon, R. D. *et al.* Palladin is an actin cross-linking protein that uses immunoglobulin-like domains to bind filamentous actin. *J Biol Chem* **283**, 6222-6231 (2008).
- 7 Upadhyay, G., Chowdhury, A. H., Vaidyanathan, B., Kim, D. & Saleque, S. Antagonistic actions of Rcor proteins regulate LSD1 activity and cellular differentiation. *Proc Natl Acad Sci U S A* **111**, 8071-8076 (2014).
- 8 Uphoff, C. C. & Drexler, H. G. Detection of mycoplasma contaminations. *Methods Mol Biol* **946**, 1-13 (2013).
- 9 Cao, Y., Charisi, A., Cheng, L. C., Jiang, T. & Girke, T. ChemmineR: a compound mining framework for R. *Bioinformatics* **24**, 1733-1734 (2008).
- 10 McCarty, K. S., Jr. *et al.* Use of a monoclonal anti-estrogen receptor antibody in the immunohistochemical evaluation of human tumors. *Cancer Res* **46**, 4244s-4248s (1986).